Chapter 39

Case Study 2: Institutional Benchmarking

R. Heller · Date: 2026-05-09 · Last changed: 2026-10-31

39.1 Objective

Compare citation summaries, fractional credits and open-access metadata in capped publication samples from five universities. These descriptive samples do not measure total productivity or field-normalised impact.

39.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
source(here::here("R", "institutional_samples.R"))
```

39.3 Data acquisition

Institution IDs were checked against OpenAlex on 30 October 2026, including the University of São Paulo record. Each request retrieves at most 300 articles; these are samples, not output totals. A failed or empty request stops the comparison instead of silently dropping an institution.

```
institutions <- tribble(
  ~short_name,    ~oa_id,
  "MIT",          "I63966007",
  "Oxford",       "I40120149",
  "ETH Zurich",   "I35440088",
  "U São Paulo",  "I17974374",
  "U Cape Town",  "I157614274"
)
```

```
fetch_works <- function(inst_id) {
  res <- fetch_openalex(
    entity = "works",
    authorships.institutions.id = inst_id,
    from_publication_date = "2020-01-01",
    to_publication_date = "2022-12-31",
    type = "article",
    options = list(sample = 300, seed = 42)
  )
  if (!is.data.frame(res) || !nrow(res)) {
    stop(glue("No usable works returned for {inst_id}; check the
     ↪  request."))
  }
  res
}

inst_data <- institutions |>
  mutate(works = map(oa_id, fetch_works))
```

39.4 Publication volume

```
inst_data |>
  mutate(n = map_int(works, nrow)) |>
  ggplot(aes(x = n, y = reorder(short_name, n))) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Sampled articles", y = NULL) +
  theme_sci()
```

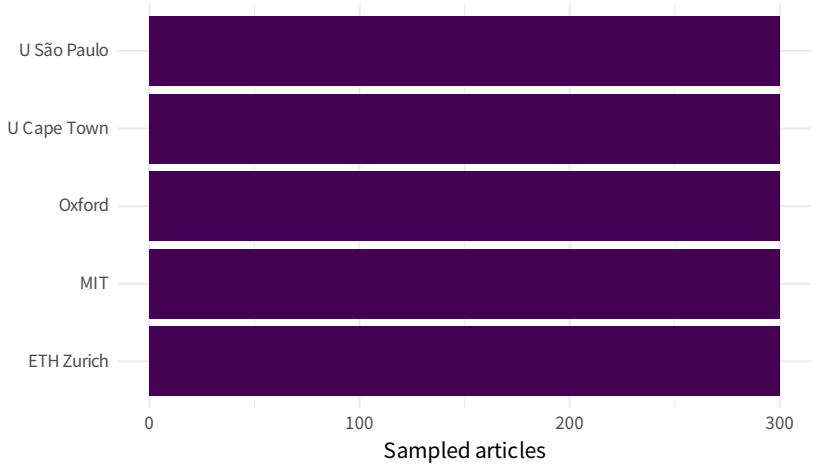


Figure 39.1: Sampled publication counts by institution.

39.5 Citation summaries

As in Chapter 13, the citation ratio uses a common year reference from the union of the five samples, de-duplicated by work ID. It adjusts for year only, not field mix or a common citation window. Missing citation counts and undefined ratios are excluded with their usable denominators shown.

```
citation_reference <- sci_pooled_year_reference(inst_data$works)
impact <- inst_data |>
  mutate(summary = map(works, sci_sample_citation_summary,
                       reference = citation_reference)) |>
  unnest(summary) |>
  select(short_name, n_sampled, n_citation_observed, mean_cites,
         median_cites, n_ratio_observed, pooled_year_ratio) |>
  arrange(short_name)

impact |>
  gt() |>
  fmt_number(columns = c(mean_cites, median_cites,
   ↪  pooled_year_ratio), decimals = 2)
```

short_name	n_sampled	n_citation_observed	mean_cites	median_cites	n_ratio_observed	pooled_year_ratio
ETH Zurich	300	300	52.02	23.00	300	1.43
MIT	300	300	51.59	25.00	300	1.40
Oxford	300	300	39.69	15.00	300	1.08

U Cape Town	300	300	29.54	12.00	300	0.77
U São Paulo	300	300	13.22	6.00	300	0.36

39.6 Full vs. fractional counting

Count each distinct affiliation once per work and divide its credit equally. Incomplete affiliations make the fractional total unavailable; the coverage column shows how many sampled works have usable data. This is bibliometric credit, not an estimate of each institution's contribution.

```
counting <- inst_data |>
  mutate(counts = map2(works, oa_id,
   ↪  sci_institution_sample_counts)) |>
  unnest(counts) |>
  select(short_name, full_count, affiliation_covered,
   ↪  frac_count) |>
  mutate(ratio = round(frac_count / full_count, 2))

counting |> gt()
```

short_name	full_count	affiliation_covered	frac_count	ratio
MIT	300	251	NA	NA
Oxford	300	263	NA	NA
ETH Zurich	300	271	NA	NA
U São Paulo	300	258	NA	NA
U Cape Town	300	255	NA	NA

```
counting |>
  pivot_longer(c(full_count, frac_count), names_to = "method",
   ↪  values_to = "count") |>
  mutate(method = recode(method, full_count = "Full", frac_count
   ↪  = "Fractional")) |>
  ggplot(aes(x = count, y = reorder(short_name, count), fill =
   ↪  method)) +
  geom_col(position = "dodge", width = 0.7) +
  scale_fill_manual(values = palette_sci(2)) +
  labs(x = "Count", y = NULL, fill = "Method") +
  theme_sci()
```

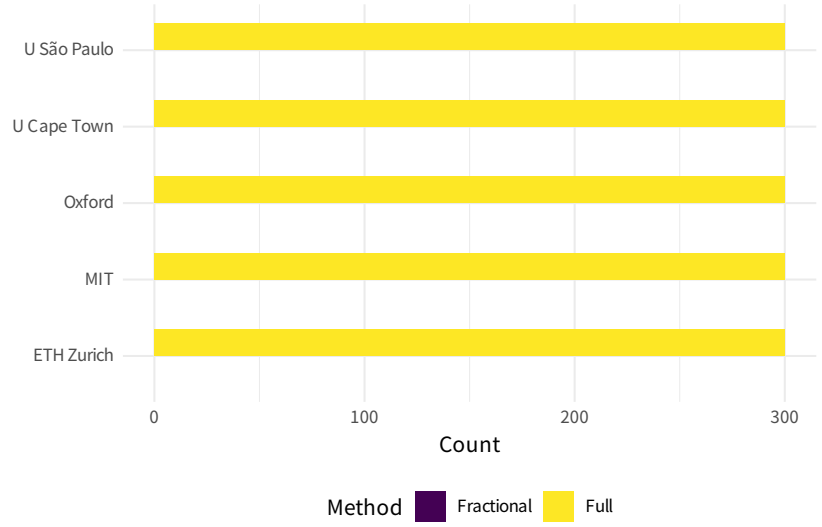


Figure 39.2: Full and fractional credits within the retrieved samples.

39.7 OA rates

```
inst_data |>
  mutate(oa_rate = map_dbl(works, \(w) mean(w$oa_status !=
   ↪  "closed", na.rm = TRUE))) |>
  ggplot(aes(x = oa_rate, y = reorder(short_name, oa_rate))) +
  geom_col(fill = palette_sci(1)) +
  scale_x_continuous(labels = scales::percent) +
  labs(x = "OA rate", y = NULL) +
  theme_sci()
```

39.8 Reading the results

1. **Sample size**: Counts are capped by the request and cannot rank institutional productivity.
2. **Citation summaries**: Means and pooled sample/year ratios describe the retrieved records. Neither is field-normalised; an h-index would also depend on sample size and is omitted here.
3. **Fractional credit**: A lower fractional/full ratio means less credit is retained under this equal-institution counting rule, subject to complete

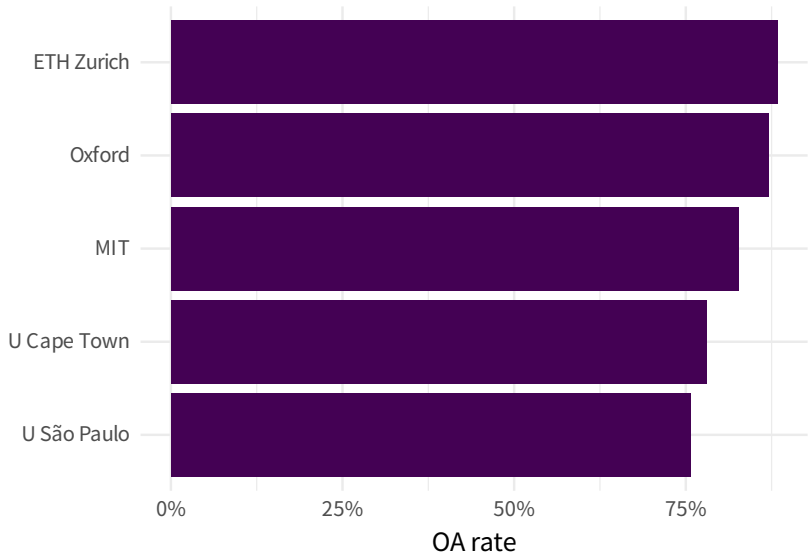


Figure 39.3: Open-access share among sampled articles with observed OA status.

affiliation coverage. It does not by itself rank the average affiliation count or collaboration quality.

4. **OA metadata**: Differences in observed OA shares do not establish the effects of national mandates or institutional policies.
5. **No single ranking**: Report coverage, sample design and indicator definitions alongside any comparison (Hicks et al., 2015).

39.9 Lessons learned

- Treat these outputs as illustrative sampled comparisons, not population rankings.
- Field mix and differing citation windows confound comparisons.
- Responsible benchmarking requires a justified reference set, transparent counting rules, missing-data coverage and uncertainty analysis.